

GEOGRAPHY

PHYSICAL GEOGRAPHY



CLIMATOLOGY

WORLD CLIMATIC TYPES

**THE TROPICAL MONSOON &
MARINE CLIMATE**

MODULE – 48

Previously seen in
Module 47

WORLD CLIMATIC TYPES

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THE TROPICAL MONSOON AND TROPICAL MARINE CLIMATES

DISTRIBUTION

1 Tropical Monsoon Climate are best developed in the Indian sub-continent, Burma, Thailand, Laos, Cambodia, parts of Vietnam and south China and northern Australia.

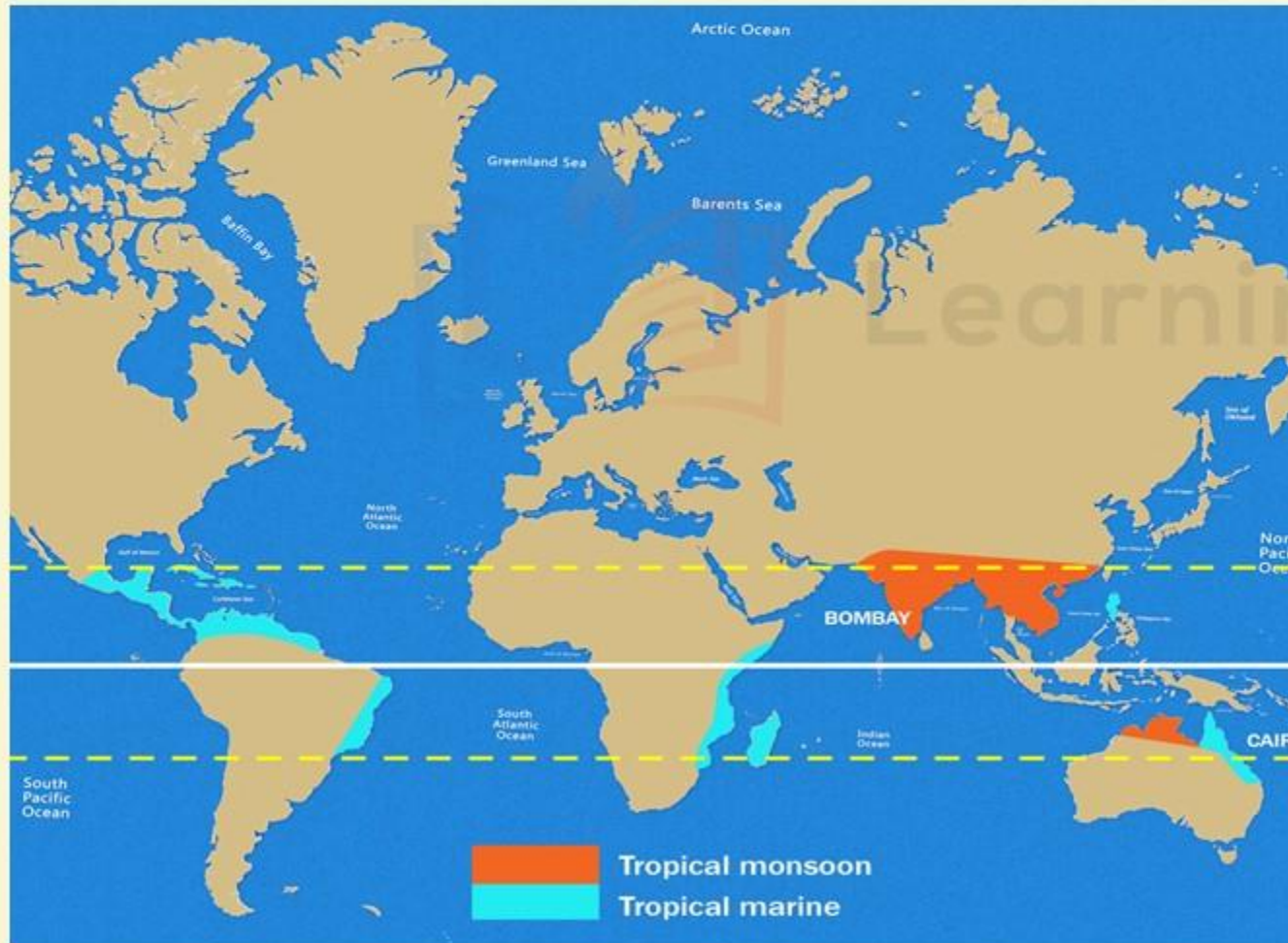
2 Outside this zone, the climate is modified by the influence of the on-shore Trade Winds all the year round and has a more evenly distributed rainfall.

3 Such a climate, better termed the Tropical Marine Climate, is experienced in Central America, West Indies, north-eastern Australia, the Philippines, parts of East Africa, Madagascar, the Guinea Coast and eastern Brazil.

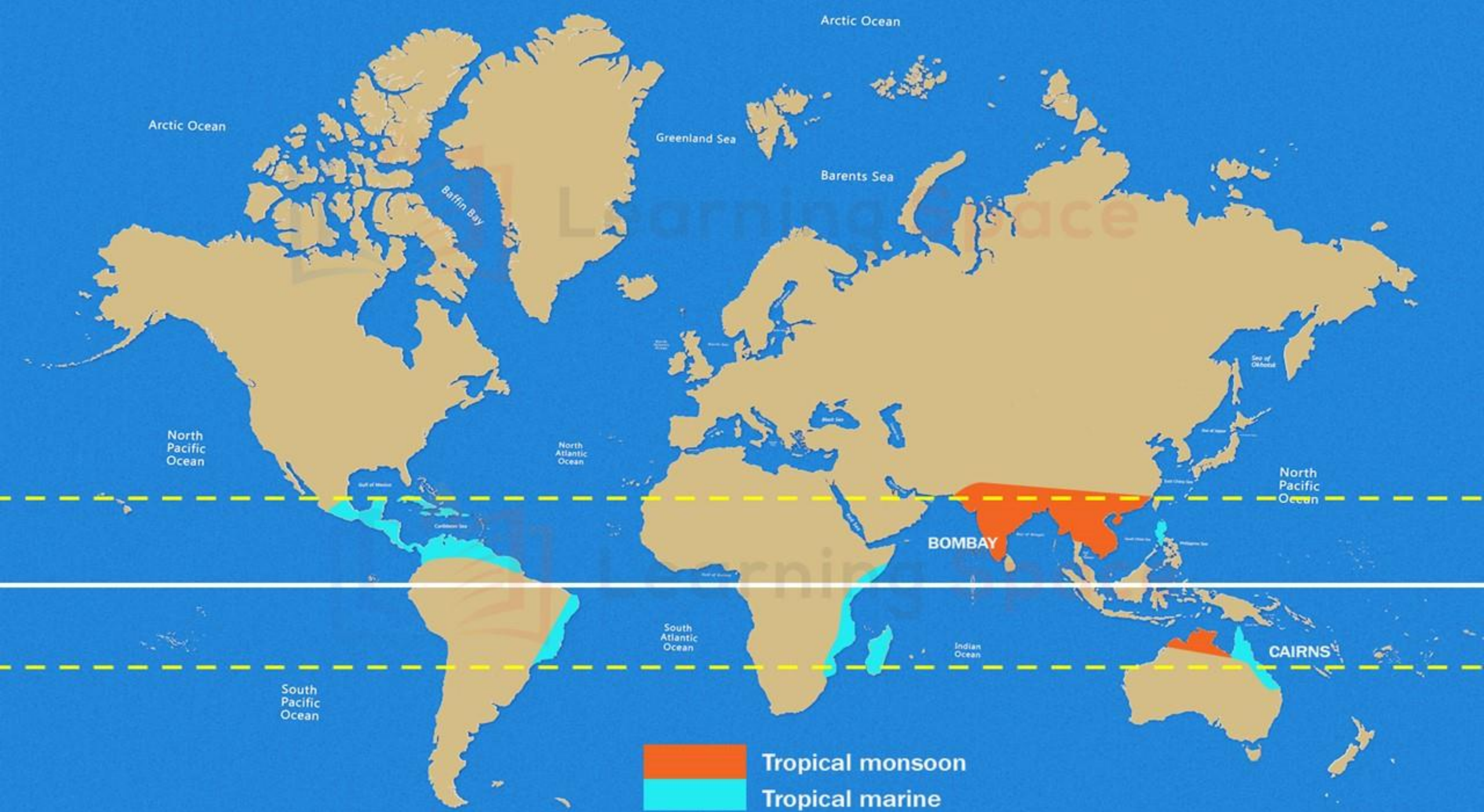
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THE TROPICAL MONSOON AND TROPICAL MARINE CLIMATES

CLIMATIC CONDITIONS IN TROPICAL MARINE LANDS



- 1 This type of climate is experienced along the eastern coasts of tropical lands, receiving steady rainfall from the Trade Winds all the time.
- 2 The rainfall is both orographic where the moist trades meet upland masses as in eastern Brazil, and convectional due to intense heating during the day and in summer.
- 3 Its tendency is towards a summer maximum as in monsoon lands, but without any distinct dry period.



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THE TROPICAL MONSOON AND
TROPICAL MARINE CLIMATES

TROPICAL MONSOON FORESTS

- 1 The natural vegetation of tropical monsoon lands depends on the amount of the summer rainfall.
- 2 Trees are normally deciduous, because of the marked dry period, during which they shed their leaves to withstand the drought.
- 3 Where the rainfall is heavy, e.g. in southern Burma, peninsular India, northern Australia and coastal regions with a tropical marine climate, the resultant vegetation is thick forest.
- 4 The forest are more open and less luxuriant than the equatorial jungle and there are far fewer species.
- 5 Most of the forest yield valuable timber, and are prized for their durable hardwood. Amongst these teak is the best known.
- 6 Burma alone accounts for as much as three-quarters of the world's production. It is such a durable timber that it is extensively used for ship building, furniture and other constructional purposes.

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THE TROPICAL MONSOON AND TROPICAL MARINE CLIMATES

TROPICAL MONSOON FORESTS



Sal tree

7 Other kinds of timber include **sal**, **acacia** and some varieties of **eucalyptus** in northern Australia.

8 Together with the forests are **bamboo thickets**, which often grow to great heights.

9 With a decrease in rainfall in summer, **the forests thin out** into **thorny scrubland** or **savanna** with **scattered trees** and **tall grass**.

10 Monsoonal Vegetation is thus most varied, ranging from forests to thickets, and from savanna to scrubland.

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THE TROPICAL MONSOON AND TROPICAL MARINE CLIMATES

AGRICULTURAL DEVELOPMENT IN THE MONSOON LANDS



- 1 Much of the monsoon forest has been cleared for agriculture to support the very dense population.
- 2 The plains are ploughed, and the hills are terraced to provide farmland.
- 3 Farms are small and the people are forever “Land Hungry”.
- 4 In their quest for land, they, have removed the natural vegetation, sometimes wantonly resulting acute soil erosion. This problem is particularly serious in the Indian sub-continent.

AGRICULTURAL DEVELOPMENT IN THE MONSOON LANDS

TYPES OF AGRICULTURE

1

WET PADI CULTIVATION

1 Rice is the most important staple crop and is grown in tropical lowlands wherever the rain exceeds 70 inches.

2 It is perhaps the most characteristic crop of the monsoon lands.

3 Irrigation water from rivers, canals, dams or wells is extensively used in the major rice producing countries.

4 Other food crops like maize, millet, sorghum, wheat, gram and beans are of subsidiary importance.

5 They are cultivated in the drier or cooler areas where rice cannot be grown.

AGRICULTURAL DEVELOPMENT IN THE MONSOON LANDS

TYPES OF AGRICULTURE

2

LOWLAND CASH CROPS



1 A wide range of lowland tropical cash crops are cultivated for the export market, after local needs have been met.

2 The most important crop in this category is cane sugar.

3 As much as two-thirds of world's sugar production comes from tropical countries.

4 Sugar is either grown on plantations or on small holdings wherever rainfall and sunshine are abundant. Some of the major producers include India, Java, Formosa, Cuba, Jamaica, Trinidad and Barbados.

AGRICULTURAL DEVELOPMENT IN THE MONSOON LANDS

TYPES OF AGRICULTURE

2

LOWLAND CASH CROPS



Jute cultivation in Gangetic Delta

5

Jute is confined almost entirely to the Ganges-Brahmaputra delta, in India and Bangladesh.

6

It has long been a leading hard fiber for the manufacture of sacks (gunny).

7

Manila hemp (abaca) is a product of the Philippines, particularly of Mindanao. It is used to make high quality rope.

AGRICULTURAL DEVELOPMENT IN THE MONSOON LANDS

TYPES OF AGRICULTURE

3 HIGHLAND PLANTATION CROPS

1 The colonization of tropical lands by Europeans gave rise to a **new form of cultivated landscape** in the cooler monsoonal highlands.

2 This is the cultivation of certain crops in tropical plantations.

3 Thousands of acres of tropical upland forests were cleared to make way for plantation agriculture in which **tea and coffee** are the most important crops.

4 Coffee originated in Ethiopia and Arabia, where it is still grown, but **Brazil now accounts for almost half the world's production of coffee**. It is mainly grown on the eastern slopes of the Brazilian plateau.

5 The crop is also cultivated on the **highland slopes in the Central American States, India and Eastern Java**.

AGRICULTURAL DEVELOPMENT IN THE MONSOON LANDS

TYPES OF AGRICULTURE

3 HIGHLAND PLANTATION CROPS



Tea plantation

6 Tea originated in China and is still an important crop there, but **as it requires moderate temperatures** (about 60⁰ F), **heavy rainfall** (over 60 inches) and **well drained highland slopes** it thrives well in the tropical monsoon zone, but preferably at a higher altitude.

7 The best regions are thus the **Himalayan foothills of India and Bangladesh**, the **Central highlands of Sri Lanka and western Java**, from all of which it is exported. In China tea is grown mostly for local consumption.

AGRICULTURAL DEVELOPMENT IN THE MONSOON LANDS

TYPES OF AGRICULTURE

4

LUMBERING



Teakwood of Burma

- 1 Wherever there are tropical forests which still have not been felled to make way for the plough, lumbering is undertaken in the more accessible areas.
- 2 This is particularly important in continental South-East Asia.
- 3 Of the tropical deciduous trees, teak, of which Burma is the leading producer, is perhaps the most sought after.

AGRICULTURAL DEVELOPMENT IN THE MONSOON LANDS

TYPES OF AGRICULTURE

4

LUMBERING



4 It is valuable on account of its great durability, strength, immunity to shrinkage, fungus attack and insects.

5 Green teak logs are so heavy that they will not float readily on water.

6 It is therefore necessary to “poison” the tree several years before actual felling, so that it is dry and light enough to be floated down the Chindwin and Irrawaddy to reach the saw mills at Rangoon.

AGRICULTURAL DEVELOPMENT IN THE MONSOON LANDS

TYPES OF AGRICULTURE

5

SHIFTING CULTIVATION



Shifting Cultivation

- 1 This most **primitive form of farming** is widely practiced here.
- 2 Instead of rotating the crops in the same field to preserve fertility, the tribesmen move to a new clearing when their first field is exhausted.
- 3 The clearing, or field, in the midst of the jungle is usually made by fire, which destroys practically everything in its way.
- 4 After planting, little attention is paid to the field either in weeding or manuring.

AGRICULTURAL DEVELOPMENT IN THE MONSOON LANDS

TYPES OF AGRICULTURE

5

SHIFTING CULTIVATION

6 The crops are left entirely to the care of nature. Maize or corn, dry padi, yams, tapioca, sweet potatoes and some beans are the most common crops.

7 Farming is entirely for subsistence i.e. everything is consumed by the farmer's family, it is mostly not traded or sold.

8 Shifting cultivation is so widely practiced amongst indigenous peoples that different local names are used in different countries. For example,

- *ladang* in Malaysia,
- *taungya* in Burma,
- *tamrai* in Thailand,
- *caingin* in the Philippines,
- *humah* in Java,
- *chena* in Sri Lanka and
- *milpa* in Africa and Central America.

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GLOBAL CLIMATE CLASSIFICATION

CLIMATE ZONE	LATITUDE	CLIMATIC TYPE
Equatorial zone	0-10° N & S	1. Hot, Wet Equatorial climate
Hot Zone	10°-30° N & S	2. a) Tropical Monsoon b) Tropical Marine
		3. Sudan type
		4. Desert type a) Saharan type b) Mid-Latitude type
Warm Temperature Zone	30°-40° N & S	5. Western Margin Mediterranean Type
		6. Central Continental Steppe type.
		7. Eastern Margin a) China type b) Gulf type C) Natal type

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GLOBAL CLIMATE CLASSIFICATION

CLIMATE ZONE	LATITUDE	CLIMATIC TYPE
Cool Temperature Zone	45°-65° N & S	8. Western Margin - British type.
		9. Siberian type - Central Continental
		10. Laurentian type - Eastern Margin
Cold Zone	65°-90° N & S	11. Arctic or Polar or Tundra climate.

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